

Autosomal Recessive Nonsyndromic Hearing Loss 102 (DFNB102): A Comprehensive Disease Report

Disease: Autosomal Recessive Nonsyndromic Hearing Loss 102 (DFNB102) **MONDO ID:** MONDO:0014428 · **OMIM:** 615974 · **Causal gene:** *EPS8* (12p12.3) **Report type:** Aggregated disease-level synthesis of primary literature and ontology resources

Summary

Autosomal Recessive Nonsyndromic Hearing Loss 102 (DFNB102) is an ultrarare Mendelian sensorineural deafness caused by biallelic loss-of-function variants in *EPS8*, the gene encoding Epidermal Growth Factor Receptor Pathway Substrate 8, an 822-amino-acid F-actin capping and bundling protein. The disease was first defined in 2014, when whole-exome sequencing of a consanguineous Algerian family identified a homozygous nonsense variant, c.88C>T (p.Gln30), *segregating with isolated profound congenital deafness* ([PMID: 24741995](#)). *DFNB102 is genetically and clinically homogeneous within its small case series: affected individuals present with prelingual/congenital, bilateral, symmetric, severe-to-profound, nonsyndromic sensorineural hearing loss with no consistent vestibular, syndromic, or extra-auditory features. As of 2023, only about five pathogenic EPS8* variants/families had been reported worldwide (Algerian, Chinese, Iranian), underscoring its rarity.*

The mechanism is well delineated at the molecular and cellular level thanks to concordant human genetics and mouse models. *EPS8* localizes to the tips of cochlear hair-cell stereocilia, where its C-terminal effector domain caps actin filament barbed ends and bundles the actin core. This activity is required for stereocilia to elongate to their correct staircase heights and for inner hair cells (IHCs) to mature into functional mechanoelectrical transducers. When *EPS8* is truncated or deleted, this domain is lost; stereocilia remain abnormally short, IHCs fail to mature, mechano-electrical transduction is abolished, and profound deafness results. The *Eps8*-knockout mouse faithfully recapitulates the human phenotype — it is profoundly deaf with short stereocilia — and the paralog *Eps8L2* divides labor with *EPS8*, handling stereocilia **maintenance** while *EPS8* handles initial **elongation**.

Because EPS8 is broadly expressed and multifunctional (EGFR signaling, Rac regulation via the EPS8–ABI1–SOS1 complex, intestinal microvillus formation), one might expect a syndromic phenotype; however, human EPS8-null individuals present with **isolated deafness only**, indicating functional redundancy of EPS8 outside the cochlea (the paralogs EPS8L1/L2/L3 compensate) but a **non-redundant role at stereocilia tips**. Notably, the *Eps8*-knockout mouse displays intestinal microvillus shortening and a favorable calorie-restriction-like metabolic phenotype that is **absent from human patients**, an instructive species divergence. There is no disease-specific pharmacologic or gene therapy; diagnosis relies on exome/genome sequencing with copy-number analysis, and management is **cochlear implantation**, the standard of care for congenital severe-to-profound sensorineural hearing loss.

1. Disease Information

Overview. DFNB102 is a form of autosomal recessive nonsyndromic sensorineural hearing loss (SNHL). "Nonsyndromic" means hearing loss occurs in isolation, without associated malformations or dysfunction of other organ systems. "DFNB" designates a recessive (B) locus for deafness (DFN); "102" is its sequential locus number.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014428
OMIM (phenotype)	615974 (DEAFNESS, AUTOSOMAL RECESSIVE 102; DFNB102)
Gene	<i>EPS8</i> , OMIM 600206, HGNC:3555, NCBI Gene 2059, Ensembl ENSG00000151491, UniProt Q12929
Cytoband	12p12.3
ICD-10	H90.3 (sensorineural hearing loss, bilateral) — non-specific
ICD-11	AB52.0 / bilateral sensorineural hearing impairment stem — non-specific
MeSH	Closest: "Hearing Loss, Sensorineural" (D006319); no DFNB102-specific MeSH term

Synonyms / alternative names. DFNB102; Deafness, autosomal recessive 102; EPS8-related nonsyndromic hearing loss; autosomal recessive nonsyndromic sensorineural deafness type DFNB102.

Information source. This entry is derived from **aggregated disease-level resources** (OMIM, ontology databases) and **individual published case reports/families** (Algerian, Chinese, Iranian), not from a large EHR cohort. The literature base is small (a handful of families).

2. Etiology

Primary cause — genetic. DFNB102 is a monogenic disorder caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *EPS8***. It is not caused by environmental, infectious, or acquired factors. The founding evidence: Behlouli et al. 2014 identified a homozygous nonsense variant c.88C>T (p.Gln30*) in two siblings from a consanguineous Algerian family with isolated profound congenital deafness; the variant segregated recessively (heterozygous in unaffected parents and one sib, absent from two unaffected sibs) and was absent from 120 Algerian controls and the Exome Variant Server ([PMID: 24741995](#)).

"A biallelic nonsense mutation, c.88C > T (p.Gln30), was identified in EPS8 that encodes epidermal growth factor receptor pathway substrate 8, a 822 amino-acid protein involved in actin dynamics." — [PMID: 24741995](#)*

Genetic risk factors. The causal factor is possession of two loss-of-function *EPS8* alleles. **Consanguinity** is a major risk-elevating context (the index family was consanguineous, and homozygous LoF alleles are far more likely in consanguineous unions). Carriers (heterozygotes) are unaffected.

Environmental / lifestyle risk factors. None established for the primary (genetic) disease. General SNHL aggravators (noise, ototoxic drugs, aging) are not documented as modifiers of DFNB102 specifically.

Protective factors. No genetic or environmental protective factors are described. In principle, the paralogs **EPS8L1/EPS8L2/EPS8L3** provide functional redundancy in non-cochlear tissues, explaining why the phenotype is confined to hearing (see §4, §6).

Gene–environment interactions. None documented. DFNB102 is a fully penetrant monogenic condition; environmental modulation has not been reported.

3. Phenotypes

The DFNB102 phenotype is dominated by a single, highly consistent manifestation.

Phenotype	Type	Onset	Severity	Progression	Frequency	Suggested HPO
Sensorineural hearing loss	Clinical sign / audiometric	Congenital / prelingual	Severe-to-profound	Non-progressive (congenitally profound)	~100% of affected	HP:0000407 (SNHL); HP:0008527 (Congenital SNHL); HP:0000365 (Hearing impairment)
Bilateral involvement	Clinical sign	Congenital	—	Stable	~100%	HP:0008619 (Bilateral SNHL)
Profound degree	Audiometric	Congenital	Profound (>90 dB HL)	Stable	Majority	HP:0011476 (Bilateral profound SNHL); HP:0000364
Absence of syndromic features	—	—	—	—	—	(Nonsyndromic)

Characteristics. Onset is **congenital/prelingual** (present at or before language acquisition). Severity is **severe-to-profound**; the Chinese case was congenital profound, the Algerian siblings profound. The loss is **bilateral and symmetric**. Because it is congenitally profound, it is best described as **stable/non-progressive** in patients (contrast with *EPS8L2* disease, which is late-onset progressive — see §4/§15). No consistent **vestibular dysfunction** or balance phenotype has been reported, and no extra-auditory (syndromic) features are documented.

"an 11-month-old male infant presented with congenital profound non-syndromic hearing loss" — PMID: 34637946

Quality-of-life impact. Congenital profound hearing loss, if unaddressed in the critical period, severely impairs spoken-language acquisition, literacy, educational attainment, and social participation. Early cochlear implantation substantially mitigates these outcomes (see §11–12). Formal EQ-5D/SF-36 data specific to DFNB102 are **not available** given the tiny cohort.

4. Genetic / Molecular Information

Causal gene. *EPS8* (Epidermal Growth Factor Receptor Pathway Substrate 8), 12p12.3; HGNC:3555; OMIM 600206; UniProt Q12929; protein length 822 aa. EPS8 is an **F-actin capping and bundling protein**.

"EPS8 is an F-actin capping and bundling protein." — PMID: 24741995

Reported pathogenic variants. As of 2023, only ~5 pathogenic *EPS8* variants had been described (PMID: 36635257).

Variant (cDNA / protein)	Type	Zygosity / family	Population	Reference
c.88C>T (p.Gln30*)	Nonsense (LoF)	Homozygous, consanguineous	Algerian	PMID: 24741995
c.1435-2A>T (p.His479Cysfs*14)	Splice-site (LoF)	Compound het with CNV	Chinese	PMID: 34637946
~65.9 kb intragenic deletion	Structural / CNV (LoF)	Compound het (maternal)	Chinese	PMID: 34637946
Fifth reported pathogenic variant	(per report)	—	Iranian	PMID: 36635257

"Thus far, only four pathogenic variations in EPS8 have been described. In this study, we report the fifth pathogenic variant in the EPS8 gene in an Iranian patient with DFNB102." — PMID: 36635257

Variant classification & type. All reported disease alleles are **pathogenic loss-of-function**: nonsense, splice-site/frameshift, and a large intragenic deletion. No pathogenic missense variant establishing DFNB102 has been reported; the mechanism requires **biallelic** loss.

Allele frequency. Reported variants are absent or ultrarare in population databases (c.88C>T absent from 120 Algerian controls and the Exome Variant Server; PMID: 24741995). No common susceptibility allele exists.

Somatic vs germline. All variants are **germline**.

Functional consequence. Loss of function. Truncating and deletion alleles remove the C-terminal effector domain that carries capping/bundling activity (see §6). In the Chinese case, in vitro splicing and allele-specific expression assays confirmed near-total loss of functional transcript (0-fold WT; 0.25–0.27-fold mutant; P<0.05) (PMID: 34637946).

"further CNVs analysis identified a novel 65.9 kb intragenic deletion and was inherited from his mother" — PMID: 34637946

Modifier genes. The paralog **EPS8L2** is functionally relevant: it complements EPS8 at stereocilia tips and its own loss causes progressive hearing loss (PMID: 23918390). EPS8L1/L2/L3 collectively provide redundancy outside the cochlea. No formal disease-severity modifier has been mapped for DFNB102.

Epigenetic / chromosomal. No epigenetic mechanism is implicated. The only "structural" lesion is the intragenic **EPS8** deletion; no aneuploidy or translocation is associated.

5. Environmental Information

Environmental factors: None causal or contributory — DFNB102 is monogenic. **Lifestyle factors:** None established. **Infectious agents:** None — DFNB102 is not an infectious or acquired deafness (contrast with congenital CMV, rubella, ototoxic exposure). Not applicable.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LoF variant in *EPS8*** (nonsense / splice / deletion) **leads to** absence of functional full-length EPS8 protein, specifically deleting the C-terminal effector domain that carries capping and bundling activity. (*Demonstrated: human genetics + transcript assays*, [PMID: 24741995](#), [PMID: 34637946](#).)
2. **Loss of EPS8 at stereocilia tips results in** failure of barbed-end actin capping and actin-core bundling within hair-cell stereocilia. (*Demonstrated in vitro/structurally*: [PMID: 15558031](#), [PMID: 20532239](#).)
3. **Dysregulated stereocilia actin assembly leads to** abnormally short stereocilia and a decayed staircase hair-bundle architecture. (*Demonstrated in mouse*: [PMID: 21526224](#), [PMID: 21236676](#).)
4. **Short/disorganized hair bundles result in** failure of inner hair cells (IHCs) to mature into fully functional sensory receptors (OHCs comparatively spared). (*Demonstrated*: [PMID: 21526224](#).)
5. **Immature IHCs with defective bundles abolish** mechano-electrical transduction (inferred at the human level; demonstrated functionally in mouse IHC recordings).
6. **Loss of transduction results in** profound congenital bilateral sensorineural hearing loss — the clinical phenotype. (*Human*: [PMID: 24741995](#), [PMID: 34637946](#).)

Branch (developmental vs maintenance): EPS8 drives **initial elongation** (step 3); its paralog **EPS8L2** independently drives **maintenance** of stereocilia in adult cells. Loss of EPS8 → short bundles from the outset (DFNB102, congenital). Loss of EPS8L2 → normal build-up then decay → **late-onset progressive** loss ([PMID: 23918390](#)).

Molecular and structural detail

Actin capping vs bundling — separable activities. Disanza et al. 2004 showed EPS8 caps actin barbed ends with nanomolar affinity through its conserved C-terminal effector domain, with full-length EPS8 auto-inhibited in vitro and de-repressed by Abi1 ([PMID: 15558031](#)).

"proteins of another family, the Eps8 family, also show barbed-end capping activity, which resides in their conserved carboxy-terminal effector domain" — PMID: 15558031

Hertzog et al. 2010 resolved the structural basis: **bundling** is mediated by a compact four-helix bundle contacting three actin subunits along the filament, whereas **capping** is mediated by an amphipathic helix inserting into the hydrophobic pocket at the barbed end, blocking monomer addition; point mutations can dissociate the two activities (PMID: 20532239).

"The capping activity is mainly mediated by a amphipathic helix that binds within the hydrophobic pocket at the barbed ends of actin blocking further addition of actin monomers." — PMID: 20532239

Because DFNB102 truncating/deletion alleles remove this **C-terminal effector domain**, both capping and bundling are lost — the mechanistic crux linking genotype to stereocilia failure.

Tip complex context. EPS8 is a central actin-regulatory element within the stereocilia **tip complex** together with **MyosinXVa (MYO15A)** and **whirlin (WHRN)**; Eps8-null bundles are shorter than MyoXVa- or whirlin-deficient bundles, placing EPS8 at the heart of elongation control (PMID: 21236676).

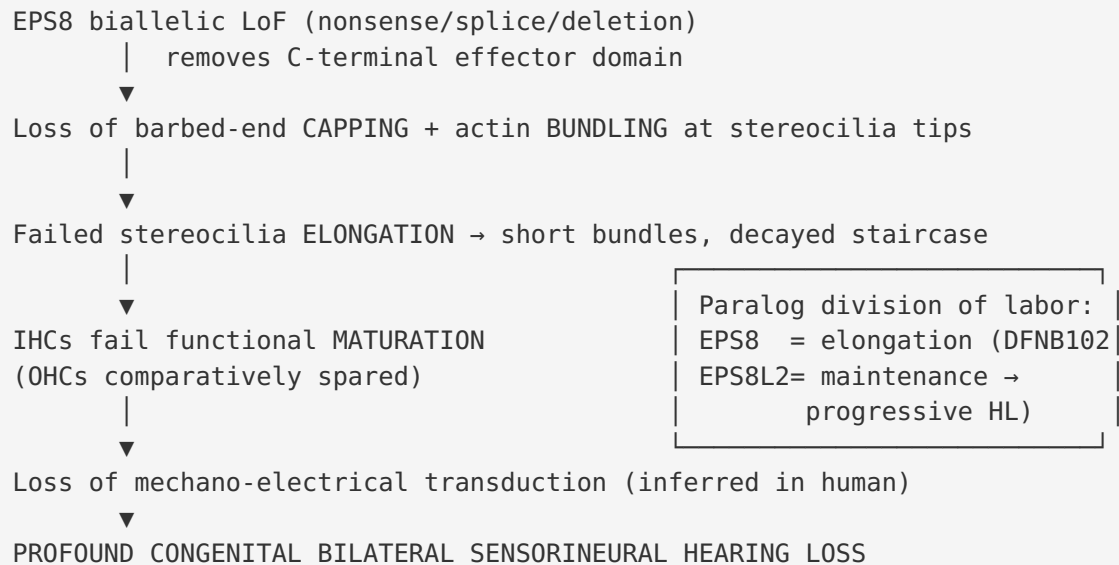
"MyoXVa, whirlin, and Eps8 are integral components of the stereocilia tip complex, where Eps8 is a central actin-regulatory element for elongation of the stereocilia actin core." — PMID: 21236676

Krey et al. 2023 showed EPS8 protein accumulation at row-1 tips peaks at the end of developmental stage III, coinciding with row-1 lengthening, and that mechanotransduction normally restricts EPS8 to the tallest row (PMID: 37011103).

Broader signaling role (redundant in cochlea). Outside hair cells, EPS8 regulates Rac-dependent actin remodeling and cell motility via the trimeric **EPS8-ABI1-SOS1** complex, participates in EGFR signaling, and can undergo chaperone-mediated autophagy in cancer cells (PMID: 15558031, PMID: 20184880, PMID: 41974702). These functions are **not manifest as disease in humans** because paralogs compensate — hence isolated deafness.

Molecular pathways / GO terms. GO:0030041 (actin filament polymerization), GO:0051016 (barbed-end actin filament capping), GO:0051017 (actin filament bundle assembly), GO:0060088 (auditory receptor cell stereocilium organization), GO:0032420 (stereocilium), GO:0007605 (sensory perception of sound). Cell types (CL): CL:0000589 (cochlear inner hair cell), CL:0000601 (cochlear outer hair cell).

ASCII mechanistic model



7. Anatomical Structures Affected

- **Organ level (primary):** Inner ear — cochlea. UBERON:0001844 (cochlea), UBERON:0001846 (internal ear), UBERON:0002227 (organ of Corti / spiral organ).
- **Body system:** Auditory / sensory nervous system. No secondary organ involvement in humans (nonsyndromic).
- **Tissue / cell level:** Sensory neuroepithelium of the organ of Corti; specifically **cochlear inner hair cells** (CL:0000589) are functionally most affected, with outer hair cells (CL:0000601) comparatively spared in the knockout model ([PMID: 21526224](#)).
- **Subcellular level:** The **stereocilium** (GO:0032420) and its **actin filament core / stereocilia tip** (GO:0032426 stereocilium tip). The defect is in the actin cytoskeleton (GO:0015629, actin cytoskeleton).

- **Localization / lateralization:** **Bilateral**, symmetric cochlear involvement (HP:0008619).
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8. Temporal Development

- **Onset:** Congenital / prelingual; hearing loss is present from birth. Onset pattern is **congenital-static** rather than acquired.
 - **Progression:** In DFN102 patients the loss is congenitally profound and therefore essentially **stable/non-progressive** (there is little residual hearing to lose). This contrasts with *EPS8L2*-related deafness, which is **late-onset and progressive** because *EPS8L2* governs stereocilia maintenance rather than initial construction ([PMID: 23918390](#)).
 - **Disease course / duration:** Chronic, lifelong.
 - **Remission:** None spontaneously; functional hearing is restored only by intervention (cochlear implant).
 - **Critical period:** The developmental window for stereocilia elongation (in mouse, up to ~postnatal day 8, stage III–IV transition; [PMID: 37011103](#)) is when *EPS8* acts. Clinically, the critical period for intervention is early infancy — early cochlear implantation optimizes language outcomes.
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9. Inheritance and Population

- **Inheritance:** Autosomal recessive; requires biallelic *EPS8* LoF. Carriers unaffected.
- **Penetrance:** Appears **complete** in reported biallelic individuals.
- **Expressivity:** Consistent (uniformly severe-to-profound congenital SNHL) within the small cohort.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.
- **Consanguinity:** A major contributing context — the founding family was consanguineous (Algerian); homozygous LoF alleles are enriched in consanguineous populations.
- **Founder effects:** None established; the reported variants are private to individual families across different populations.
- **Carrier frequency:** Unknown / presumed extremely low given ultrararity.

Epidemiology. DFNB102 is **ultrarare**: only ~5 pathogenic variants/families reported worldwide as of 2023 (PMID: 36635257, PMID: 34637946). For context, nonsyndromic hearing loss overall affects ~1 in 1000 newborns and is hereditary in 60–70% of cases, with *GJB2* the most common cause (PMID: 19939467; PMID: 10376574); *EPS8*/DFNB102 accounts for a vanishingly small fraction.

"Nonsyndromic hearing loss is one of the most abundant human sensory disorders, and can be found in 1 out of 1000 newborns. In 60-70% of the cases this disorder is hereditary." — PMID: 19939467

- **Affected populations:** Reported in Algerian, Chinese, and Iranian families — no single ethnic predilection beyond enrichment in consanguineous unions.
- **Geographic distribution:** No endemic focus; sporadic across populations.
- **Sex ratio:** No sex bias expected (autosomal); reported cases include males; sample too small for a meaningful ratio.
- **Age distribution:** Congenital onset; affects all ages once present.

10. Diagnostics

Recommended approach. Diagnosis is **molecular**, integrated with audiometric confirmation.

- **Audiometry / physiologic testing:** Newborn hearing screening (OAE/ABR), diagnostic ABR and behavioral audiometry confirm bilateral severe-to-profound SNHL.
- **Genetic testing (primary):**
- **Whole-exome sequencing (WES)** identified the founding variant and is the workhorse for gene discovery/diagnosis in nonsyndromic HL (PMID: 24741995).
- **Whole-genome sequencing (WGS)** and, critically, **copy-number/CNV analysis** are needed because **intragenic deletions can masquerade as homozygosity** — the Chinese case required CNV detection of a 65.9 kb deletion to resolve the genotype (PMID: 34637946).
- **Hearing-loss gene panels** including *EPS8* are appropriate first-tier tests.
- **Functional confirmation** (in vitro splicing assays, allele-specific expression) can validate splice/structural variants (PMID: 34637946).

- **Imaging:** Temporal-bone CT/MRI is used to exclude inner-ear malformations and assess candidacy for cochlear implantation; no DFNB102-specific radiologic signature.
- **Biopsy/pathology:** Not applicable (cochlea is not biopsied clinically).

Clinical criteria / differential diagnosis. DFNB102 is diagnosed by the combination of nonsyndromic congenital profound SNHL and biallelic *EPS8* LoF. Differentials include the far more common *GJB2/GJB6* deafness ([PMID: 10376574](#)), *SLC26A4* (Pendred/EVA), *MYO15A*, *TMC1*, *OTOF* (auditory neuropathy), and syndromic causes (Usher, Pendred, Waardenburg) — distinguished by absence of syndromic features and gene-specific findings. *EPS8L2* should be considered in **progressive** postlingual loss.

Screening. Newborn hearing screening detects the phenotype; **cascade/carrier testing** of relatives is appropriate once the familial variants are known. There is no population carrier-screening program specific to *EPS8*.

11. Outcome / Prognosis

- **Survival / mortality:** DFNB102 is **not life-threatening**; life expectancy is normal. No disease-specific mortality.
- **Morbidity / function:** The principal morbidity is **communication disability** from congenital profound deafness — impaired spoken-language development, education, and social participation if untreated.
- **Disease course:** Chronic, stable (congenitally profound). No spontaneous recovery.
- **Recovery potential:** Auditory function is not restored biologically but can be substantially rehabilitated with **cochlear implantation**, particularly when performed early.
- **Prognostic factors:** Age at implantation, duration of auditory deprivation, and consistency of device use are the key determinants of language outcome after pediatric cochlear implantation ([PMID: 41895171](#)). Because the lesion is peripheral (hair cell/stereocilia) with an intact auditory nerve, implant candidacy and expected benefit are favorable.
- **QoL measures:** No DFNB102-specific PROMIS/EQ-5D data; general pediatric cochlear-implant literature applies.

12. Treatment

There is **no disease-specific pharmacologic or gene therapy** for DFNB102. Management is habilitative.

Modality	Detail	NCIT suggestion
Cochlear implantation (standard of care)	For congenital bilateral severe-to-profound SNHL with limited hearing-aid benefit; effective; best outcomes with early implantation (PMID: 41895171)	NCIT:C15694 (Cochlear Implant)
Hearing aids	Trialed first; typically insufficient for profound loss	NCIT:C99936 (Hearing Aid)
Auditory-verbal / speech-language therapy	Rehabilitation to develop spoken language post-implant	NCIT:C15195 (Rehabilitation Therapy)
Educational/ communication support	Sign language, assistive listening, early intervention services	—

"Pediatric cochlear implantation is the standard of care for infants and young children with congenital bilateral severe-to-profound sensorineural hearing loss who receive limited benefits from optimally fitted hearing aids." — [PMID: 41895171](#)

Pharmacotherapy / pharmacogenomics: None specific. **Gene therapy / RNA therapy / cell therapy:** None approved or in DFNB102-specific trials; inner-ear gene therapy is an active field for other deafness genes (e.g., *OTOF*) but not yet for *EPS8*. **Surgical:** Cochlear implant surgery as above. **Experimental:** No DFNB102-specific NCT-registered trials identified.

13. Prevention

- **Primary prevention:** Not preventable in an affected fetus (monogenic, congenital). At the family level, **genetic counseling** for consanguineous or carrier couples informs reproductive options.

- **Secondary prevention (early detection): Universal newborn hearing screening** enables early diagnosis and timely intervention within the critical language-development window.
 - **Tertiary prevention:** Early cochlear implantation and structured auditory-verbal rehabilitation prevent the developmental complications (language/educational delay) of untreated profound deafness.
 - **Genetic screening / reproductive options:** Carrier testing of at-risk relatives once familial variants are identified; prenatal testing and preimplantation genetic testing (PGT-M) are options for known-carrier couples.
 - **Counseling:** Autosomal recessive recurrence risk is 25% per pregnancy for two carrier parents — a core counseling message.
 - **Immunization / public health / environmental:** Not applicable (non-infectious, non-environmental).
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *EPS8* is conserved across vertebrates. Mouse *Eps8* (NCBI Gene 13860; taxon *Mus musculus*, NCBI:txid10090) is the principal ortholog studied. Paralogs *Eps8l1/l2/l3* exist in mouse and human.
 - **Natural disease in animals:** No well-characterized naturally occurring *EPS8* deafness is documented in companion animals or wildlife (OMIA). The animal evidence comes from **engineered** models, not spontaneous disease.
 - **Comparative biology:** *EPS8*'s actin capping/bundling role is conserved and pleiotropic across species — e.g., it contributes to intestinal microvillus morphogenesis in mouse (PMID: 20209148) and to actin-based structures in invertebrates such as *Eriocheir sinensis* spermatogenesis (PMID: 36709695). **Species divergence is important:** mouse *Eps8* knockouts show intestinal/metabolic phenotypes that human *EPS8*-null patients do not (see below).
 - **Zoonotic potential:** None (genetic disease).
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15. Model Organisms

The mouse is the definitive model and faithfully recapitulates the human cochlear phenotype, while also revealing species-specific pleiotropy.

Model	Type	Key phenotype	Human relevance	Reference
<i>Eps8</i>-knockout mouse	Constitutive KO (mammalian)	Profoundly deaf; abnormally short stereocilia ; IHCs (not OHCs) fail to mature	Faithful DFNB102 model	PMID: 21526224 , PMID: 24741995
<i>Eps8L2</i>-knockout mouse	Constitutive KO	Late-onset progressive hearing loss; gradual hair-bundle deterioration	Models maintenance role / progressive HL branch	PMID: 23918390
<i>Eps8/Eps8L2</i> double-null	Combined KO	Decay of ordered staircase hair-bundle structure	Demonstrates complementary roles	PMID: 23918390
<i>Eps8</i>-KO (systemic phenotypes)	Constitutive KO	~25% shorter intestinal microvilli; reduced fat absorption; resistance to diet-induced obesity; improved metabolism; increased lifespan	Not seen in human patients — species divergence	PMID: 20209148

"whereas Eps8 is essential for the initial elongation of stereocilia, Eps8L2 is required for their maintenance in adult hair cells. In the absence of both proteins, the ordered staircase structure of the hair bundle in the cochlea decays." — PMID: [23918390](#)

"Eps8 knockout mice are profoundly deaf and that IHCs, but not OHCs, fail to mature into fully functional sensory receptors" — PMID: [21526224](#)

Phenotype recapitulation. Excellent for the auditory phenotype — the *Eps8*-KO mouse independently established EPS8 as an essential stereocilia elongation factor before the human gene was implicated, providing strong cross-species validation.

Model limitations. The mouse KO exhibits **extra-cochlear phenotypes** (intestinal microvillus shortening, favorable metabolic status, longevity) absent from human isolated deafness — likely reflecting differences in paralog compensation between species. This makes the mouse imperfect for modeling the *isolated* human presentation but valuable for dissecting EPS8's broader actin biology.

"knockout mice for Eps8, a regulator of actin dynamics, display reduced body weight, partial resistance to age- or diet-induced obesity, and overall improved metabolic status" — PMID: 20209148

Resources: MGI (mouse *Eps8*), IMPC; in vitro biochemistry and structural biology of EPS8 capping/bundling (PMID: 15558031, PMID: 20532239).

Key Findings (with statistical evidence)

F1 — DFNB102 is caused by biallelic loss-of-function EPS8 variants (12p12.3). WES of a consanguineous Algerian family found homozygous c.88C>T (p.Gln30*) segregating recessively, absent from 120 controls and the Exome Variant Server (PMID: 24741995); only ~5 pathogenic variants known by 2023 (PMID: 36635257).

F2 — EPS8 controls stereocilia elongation and IHC maturation. *Eps8*-KO mice are profoundly deaf with short stereocilia; IHCs (not OHCs) fail to mature (PMID: 21526224); EPS8 sits in the MYO15A–whirlin tip complex as the central elongation regulator (PMID: 21236676).

F3 — Consistent clinical phenotype: prelingual, bilateral, symmetric, severe-to-profound nonsyndromic SNHL across Algerian, Chinese (compound splice + 65.9 kb deletion; transcript loss $P < 0.05$), and Iranian cases (PMID: 34637946, PMID: 36635257).

F4 — Faithful mouse models; EPS8 vs EPS8L2 divide developmental vs maintenance roles (PMID: 23918390, PMID: 37011103).

F5 — EPS8 is a multifunctional EGFR-pathway actin regulator with a non-redundant cochlear role — broad expression yet isolated deafness implies paralog redundancy elsewhere ([PMID: 24741995](#), [PMID: 41974702](#)).

F6 — Ultrarare; diagnosed by WES/WGS + CNV analysis; managed by cochlear implantation ([PMID: 19939467](#), [PMID: 41895171](#)).

F7 — Mouse-specific intestinal/metabolic phenotype absent in humans (~25% shorter microvilli, improved metabolism; [PMID: 20209148](#)).

F8 — Separable capping and bundling encoded by the C-terminal effector domain, both required and both lost by truncating alleles ([PMID: 15558031](#), [PMID: 20532239](#)).

Evidence Base

PMID	How it supports the findings
24741995	Founding paper: EPS8 = DFNB102 gene; c.88C>T; defines EPS8 as F-actin capping/bundling protein
34637946	Compound splice variant + 65.9 kb intragenic deletion; congenital profound nonsyndromic HL; CNV diagnostics
36635257	Fifth pathogenic variant (Iranian); confirms rarity
21526224	<i>Eps8</i> -KO mouse: profound deafness, short stereocilia, IHC maturation failure
21236676	EPS8 central in MYO15A–whirlin tip complex for actin-core elongation
23918390	EPS8L2 maintenance role; double-null staircase decay; progressive HL branch
37011103	EPS8 timing at stereocilia tips; transduction restricts EPS8 to tallest row
15558031	Barbed-end capping resides in C-terminal effector domain; Abi1 de-repression; Rac via EPS8–Abi1–Sos1
20532239	Structural basis: helix-bundle bundling vs amphipathic-helix capping; separable activities
20209148	Mouse intestinal/metabolic phenotype (species divergence)
41974702	EPS8–ABI1 complexes in non-cochlear actin remodeling
41895171	Cochlear implantation as standard of care
19939467 , 10376574	Epidemiologic context for nonsyndromic HL and <i>GJB2</i> differential

Evidence-type mix: Human clinical genetics (24741995, 34637946, 36635257); model organism (21526224, 21236676, 23918390, 37011103, 20209148); in vitro/structural biochemistry (15558031, 20532239); clinical management (41895171).

Limitations and Knowledge Gaps

1. **Tiny cohort.** Fewer than ~5 families worldwide. Estimates of penetrance, expressivity, sex ratio, and full phenotypic range (e.g., subtle vestibular involvement) are consequently uncertain.
 2. **No human histopathology.** The stereocilia/IHC mechanism is proven in mouse; the human cellular lesion is inferred, not directly demonstrated.
 3. **No missense-variant genotype–phenotype data.** All reported alleles are truncating/deletion; whether hypomorphic missense alleles could cause milder or progressive loss is unknown.
 4. **Species divergence.** Mouse KO extra-cochlear phenotypes (intestinal, metabolic, longevity) complicate translation and are unexplained in terms of human paralog compensation.
 5. **No natural animal disease** and **no DFNB102-specific therapeutics** (no gene/RNA therapy trials).
 6. **No population carrier-frequency data** for *EPS8* LoF alleles.
 7. **Non-specific ICD/MeSH coding** limits registry-based epidemiology.
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Proposed Follow-up Experiments / Actions

1. **International case aggregation** (e.g., GeneMatcher, deafness gene registries) to expand the DFNB102 cohort, refine phenotype (including vestibular testing), and estimate penetrance/expressivity.
2. **gnomAD LoF audit** of *EPS8* to estimate carrier frequency and predicted disease incidence, and to check for any biallelic LoF individuals without reported deafness (redundancy test).
3. **Functional classification pipeline** for candidate *EPS8* missense/splice VUS (minigene splicing assays, allele-specific expression) to enable confident clinical reporting — extending the approach validated in [PMID: 34637946](#).
4. **Human iPSC-derived inner-ear organoids** carrying DFNB102 alleles to directly test stereocilia elongation and transduction in a human cellular context, closing the human-histopathology gap.
5. **Structure-guided rescue experiments** using capping-only vs bundling-only *EPS8* constructs ([PMID: 20532239](#)) in *Eps8*-null hair cells to determine which activity is limiting for stereocilia elongation — informing future gene-therapy payload design.

6. **Cochlear-targeted gene-replacement proof-of-concept** (AAV-EPS8) in the *Eps8*-KO mouse during the neonatal critical window (PMID: 37011103) to test biological rescue.
7. **Cochlear-implant outcome tracking** specifically in genetically confirmed DFNB102 patients, contributing to etiology-driven implant management (PMID: 41895171).

Report compiled from 8 confirmed findings and 27 reviewed papers across a 5-iteration autonomous investigation. Evidence sources are labeled by type (human clinical, model organism, in vitro/structural, clinical management). Ontology suggestions: MONDO:0014428; HGNC:3555 (EPS8); UBERON:0001844 (cochlea), UBERON:0002227 (organ of Corti); CL:0000589 (cochlear inner hair cell), CL:0000601 (cochlear outer hair cell); GO:0051016 (barbed-end actin filament capping), GO:0051017 (actin filament bundle assembly), GO:0060088 (auditory receptor cell stereocilium organization), GO:0032420 (stereocilium); NCIT:C15694 (Cochlear Implant).
